

COASTER GRID: The Kinetic Theme Park Dispatcher

A Fusion of Mini Metro Logistics and RollerCoaster Tycoon Physics | Godot 4.x

1. Executive Vision & Core High Concept

COASTER GRID reimagines the addictive line-management triage of Mini Metro through the beloved lens of Chris Sawyer's RollerCoaster Tycoon (1999). Instead of routing abstract subway lines on a flat map, the player constructs 2.5D isometric roller coasters to ferry swarms of park visitors ("Peeps") between themed park zones. Coaster cars obey authentic momentum and gravity: lift hills build potential energy, steep drops generate extreme speed, and unbanked turns risk dangerous derailment.

2. The 2.5D Isometric Aesthetic Specification

- **Projection & Camera Angle:** Classic Dimetric 2:1 Isometric Projection (30-degree tilt, 45-degree horizontal rotation). Fixed orthographic viewport providing timeless retro readability with depth layering.
- **Visual Identity (Chris Sawyer Tribute):** Crisp grid-aligned isometric terrain tiles (Grass, Dirt, Water, Cobblestone paths), vibrant tubular steel coaster tracks with cross-ties, and tiny animated Peeps wearing colorful caps and jackets.
- **Layering & Depth Sorting:** Elevated tracks cast dynamic soft shadows onto the ground and intersecting lower tracks. Tunnels cut cleanly through elevated terrain cliffs, rendering translucent cutaways when cars plunge underground.

3. The Dual-Loop Gameplay Framework

- **Macro-Loop (Guest Triage & Park Logistics):** Guests spawn from the park main gate with distinct ride preferences. As park popularity surges, station queues backlog. If queue wait times exceed tolerance thresholds, guests rage-quit, cratering Park Rating.
- **Micro-Loop (Kinetic Coaster Engineering):** You do not draw abstract bezier curves; you snap physical modular track pieces together in 3D grid space. Every coaster must balance high Excitement (for thrill-seeking guests) against Nausea and Physics Safety (to prevent derailments).

The Core Innovation: "Transit Tycoon Triage"

Traditional Tycoon games are slow-paced sandbox builders. Mini Metro is a fast-paced panic survival game. COASTER GRID unites both: you must rapidly expand, re-route, and split your coaster lines on the fly while managing real kinetic momentum and queue bottlenecks before your park rating collapses.

Isometric Grid Geometry, Track Types, and Physical Simulation

Physical Model: Potential Energy -> Kinetic Energy -> Friction Dissipation

1. Modular Isometric Track Grid System

Track construction operates on an isometric voxel grid (X, Y, Z coordinates). Tracks snap seamlessly from grid cell to grid cell with explicit elevation changes:

- **Station Platform (Flat, Z=0..N):** Where guests board and disembark. Contains automatic pneumatic brakes and dispatch launch timers (e.g., launch every 10 seconds or when full).
- **Chain Lift Hill (Inclined Upward, +1 Z per tile):** Motorized chain pulls coaster cars upward at a fixed slow speed (15 km/h), converting mechanical power into gravitational potential energy ($E_p = m * g * h$).
- **Gravity Drop (Inclined Downward, -1 Z to -2 Z):** Steep drops convert potential energy into extreme kinetic velocity ($E_k = 0.5 * m * v^2$). Tighter drops deliver intense G-force acceleration.
- **Banked & Unbanked Curves (90° Turns):** Flat curves taken at high speeds generate deadly lateral G-forces. Banked curves tilt the car inward, preserving passenger comfort and preventing rollover.
- **Special Elements (Unlocks):** Vertical Loop (requires high threshold velocity to clear the crest without rolling backwards), Friction Brakes (bleed off excess speed), and Booster Tires (flat acceleration).

2. The Kinetic Coaster Physics Engine

- **Energy Conservation Formula:** Total Energy $E = E_p + E_k - W_{\text{friction}}$. Velocity at any point is governed by $v = \sqrt{\max(0, 2 * (E_{\text{total}} - m * g * h) / m)}$. Cars naturally stall and roll backwards if launch speed cannot overcome hill height.
- **Block Brakes & Multi-Train Security:** Coasters can run multiple trains safely only if separated by Block Brake sections. If a train stalls on track and the trailing train catches up: instant catastrophic collision!
- **Derailment & Crash Stakes:** If a coaster train enters a sharp unbanked turn or the end of an incomplete track at speeds exceeding 75 km/h, the train violently derails, exploding into confetti and instantly failing the current park level.

Coaster Ride Statistics (Dynamic Calculation)

- **Excitement Rating (Target: 6.0 - 9.0):** Driven by high speed drops, airtime hills (negative vertical Gs), and inversions.
- **Intensity Rating (Target: 4.0 - 7.0):** Increases with lateral G-forces, sudden drops, and total track length. If too high (>9.0), casual guests refuse to ride.
- **Nausea Rating (Target: < 4.0):** Driven by sustained spinning, abrupt directional changes, and excessive positive Gs. High nausea causes guests to vomit at exit paths.

Peep AI Behaviors, Station Queue Bottlenecks, and Scarcity Economy

Core Tension: High Throughput vs. Guest Comfort and Queue Capacity

1. Peep Behavioral Archetypes

Guests arrive via the park entrance in continuous waves. Each guest is a distinct AI agent simulated with cash, energy, nausea tolerance, and preferred excitement thresholds:

- **Thrill Seekers (Red Shirts, 35%):** Demand Excitement > 7.0 and Intensity > 6.0 . They actively hunt high drops and loops; if only gentle rides are available, they grow bored and depart.
- **Family / Casual Guests (Green Shirts, 45%):** Prefer Excitement $4.0 - 6.5$ and Intensity < 5.0 . They enjoy scenic elevated coasters and gentle drops. High intensity rides trigger panic and refusal.
- **Queasy Peeps (Yellow Shirts, 20%):** Extremely low Nausea tolerance. Riding high-G coasters fills their sick meter to 100%, causing them to vomit immediately upon stepping onto the exit footpath.

2. The Station Overcrowding Clock (The Mini Metro Triage)

- **Queue Capacity Limit:** Each station platform has a finite queue tile capacity (e.g., 16 guests).
- **The Impatience Warning Clock:** When the station queue fills to 100%, a circular radial warning timer appears over the station (identical to Mini Metro).
- **Queue Evacuation & Park Rating Penalty:** If the timer completes before an empty coaster train arrives to load passengers: guests riot, break fences, and storm out of the park. Park Rating plummets by 25%.
- **Failure State:** If Park Rating drops below 30% for 30 consecutive seconds, the theme park is condemned by city inspectors—Game Over.

3. The Weekly Scarcity Economy (Upgrades)

- **Every In-Game "Sunday" (90 Seconds of Play):** The clock pauses and presents the player with a critical resource grant choice:
- **Option A (Track Logistics):** +1 New Coaster Station & Color Line OR +10 Track Sections.
- **Option B (Train Capacity):** +1 Train Locomotive OR +2 Carriages per train (doubling guest boarding throughput).
- **Option C (Special Engineering):** +1 Inversion Loop Piece OR +1 Set of Pneumatic Block Brakes (allowing 2 trains on 1 track).

Park Sanitation & Footpath Maintenance

When sick guests vomit on exit paths, surrounding guests walking over the puddle experience a disgust spike, speeding up their impatience timer. Players must strategically position trash cans, benches, or hire automated Handymen to keep arterial paths spotless.

Engine Node Hierarchy, Coordinate Math, and Jam Sprint Milestones

Targeting 60 FPS Smooth Simulation & Zero Memory Leaks on Web/Desktop

1. Godot 4 Technical Architecture (2.5D Implementation)

- **Camera System (Orthographic 3D):** Use a Node3D root with Camera3D set to PROJECTION_ORTHOGONAL, rotation_degrees = Vector3(-30, 45, 0). This delivers authentic 2.5D RCT perspective while leveraging Godot's native 3D vector math.
- **Track Traversal via Path3D & PathFollow3D:** Each coaster route dynamically compiles into a Path3D with a Curve3D. Trains are PathFollow3D nodes. Speed along the curve is dynamically computed in _physics_process based on delta height.
- **Isometric Grid Coordinates:** GridManager coordinates world positions using: $\text{pos} = \text{Vector3}((x - y) * \text{TILE_WIDTH} / 2, z * \text{STEP_HEIGHT}, (x + y) * \text{TILE_DEPTH} / 2)$. Track pieces are modular MeshInstances or MultiMeshInstances for high draw-call performance.
- **Guest Navigation via AStar3D:** Park footpaths register nodes into an AStar3D graph. Peeps query paths from entrance -> station queue -> ride -> exit path -> benches.

2. 5-Day GameJam Sprint Schedule (Sept 6 – Sept 11)

- **Day 1 — Isometric Foundation & Track Placement:** Implement orthographic camera, mouse cursor snapping to isometric grid, and basic track piece placement (Straight, Curve, Incline, Station).
- **Day 2 — Kinetic Physics & Coaster Dispatch:** Generate Curve3D from placed tracks. Implement PathFollow3D train with gravitational acceleration (speed up on drops, slow on inclines, chain lift speed).
- **Day 3 — Guest Peeps & Queue Management:** Spawn Peeps from entrance gate. Pathfind to station queue lines. Implement boarding logic, train capacity, and ride dispatch.
- **Day 4 — Metrics, Overcrowding Clock & Upgrades:** Calculate Excitement/Intensity/Nausea. Implement station overcrowding radial timer, Park Rating meter, and weekly upgrade grant UI.
- **Day 5 — Audio Juice, Sound Effects & Submission Polish:** Add mechanical chain-lift clicking audio, coaster roar, crowd chimes, UI bells, vomit particles, title screen, and itch.io Web export.

Scoping Guardrail: How to Avoid Feature Creep in 5 Days

Do NOT implement complex terrain terraforming or 50 different ride types. Focus 100% on ONE coaster style (Tubular Steel Coaster), ONE park map, and tight, polished track-laying physics. A clean 3-level prototype with juicy coaster sounds will score top marks in Functionality and Technical Implementation.